

■ DevOps Practical Assignment

■ Objective

This assignment is designed to test your hands-on understanding of **Git, Open Source Contribution, Maven, and Jenkins (Master-Slave Architecture)**. You are expected to work like a real DevOps engineer, not just follow commands mechanically.

■ Task 1: Open Source Contribution (GitHub)

- Fork the repository: <https://github.com/Ankan619/HitPractice.git>
- Clone your forked repository to your local system
- Create a new branch with a meaningful name
- Make valid improvements (documentation, scripts, examples, or enhancements)
- Commit changes with proper commit messages
- Push changes to your forked repository
- Create a **Pull Request** to the original repository
- **Submission proof:** Pull Request URL

■ Task 2: Git Branching & Merging Practical

Perform the following steps exactly in the given order:

- 1 Create folder-1 and add 5 files inside it
- 2 Create folder-2 and add 5 files inside it
- 3 Create folder-3 and add 5 files inside it
- 4 Add all folders and files to master branch (empty files initially)
- 5 Create a branch named **first**
- 6 Modify file1, file2, file3 in folder-1 (first branch)
- 7 Modify file1, file2, file3 in folder-2 (first branch)
- 8 Create a branch named **second**
- 9 Modify file1, file2, file3 in folder-3 (second branch)
- 10 Switch to master and modify **all files** by adding content
- 11 Merge second branch into first branch
- 12 Merge first branch into master branch

■ Task 3: Maven Mini Game Project

Create a mini game of your choice (Console-based or GUI-based) using **Java + Maven**.

- Use Maven project structure
- Add proper **pom.xml** with dependencies
- Game should run using command line
- Generate an **Executable JAR** file
- Run the JAR using: **java -jar yourgame.jar**
- Upload source code and JAR file to GitHub

■ Task 4: Jenkins Master–Slave Architecture

Build the same Maven game project using Jenkins Master–Slave (Agent) setup.

- Install Jenkins on Master node
- Configure at least one Slave/Agent node
- Connect agent using SSH or JNLP
- Create Jenkins pipeline or freestyle job
- Configure job to run on Slave node
- Build Maven project on Slave
- Archive the generated JAR artifact
- Attach screenshots of Jenkins Master, Slave, and Build logs

■ Evaluation Criteria

- Correct Git workflow & branching
- Quality of Open Source contribution
- Successful executable JAR creation
- Proper Jenkins Master–Slave configuration
- Clean commits and documentation

■■ **Note:** This is a hands-on assignment. Copy–paste or fake screenshots will result in disqualification.